

NAME

codata_2018 - Fortran module containing the CODATA 2018 constants

LIBRARY

codata (**-libcodata, -lcodata**)

SYNOPSIS

use *codata*

include "codata.h"

import *pycodata*

DESCRIPTION

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018
- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018

- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018
- ATOMIC_UNIT_OF_PERMITTIVITY_2018
- ATOMIC_UNIT_OF_TIME_2018
- ATOMIC_UNIT_OF_VELOCITY_2018
- AVOGADRO_CONSTANT_2018
- BOHR_MAGNETON_2018
- BOHR_MAGNETON_IN_EV_T_2018
- BOHR_MAGNETON_IN_HZ_T_2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- BOHR_MAGNETON_IN_K_T_2018
- BOHR_RADIUS_2018
- BOLTZMANN_CONSTANT_2018
- BOLTZMANN_CONSTANT_IN_EV_K_2018
- BOLTZMANN_CONSTANT_IN_HZ_K_2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018
- CLASSICAL_ELECTRON_RADIUS_2018
- COMPTON_WAVELENGTH_2018
- CONDUCTANCE_QUANTUM_2018
- CONVENTIONAL_VALUE_OF_AMPERE_90_2018
- CONVENTIONAL_VALUE_OF_COULOMB_90_2018
- CONVENTIONAL_VALUE_OF_FARAD_90_2018
- CONVENTIONAL_VALUE_OF_HENRY_90_2018
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_OHM_90_2018
- CONVENTIONAL_VALUE_OF_VOLT_90_2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018

- CONVENTIONAL_VALUE_OF_WATT_90_2018
- COPPER_X_UNIT_2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2018
- DEUTERON_ELECTRON_MASS_RATIO_2018
- DEUTERON_G_FACTOR_2018
- DEUTERON_MAG_MOM_2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- DEUTERON_MASS_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- DEUTERON_MASS_IN_U_2018
- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018
- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018

- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018
- ELEMENTARY_CHARGE_2018
- ELEMENTARY_CHARGE_OVER_H_BAR_2018
- FARADAY_CONSTANT_2018
- FERMI_COUPLING_CONSTANT_2018
- FINE_STRUCTURE_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018
- HARTREE_ENERGY_2018
- HARTREE_ENERGY_IN_EV_2018
- HARTREE_HERTZ_RELATIONSHIP_2018
- HARTREE_INVERSE_METER_RELATIONSHIP_2018
- HARTREE_JOULE_RELATIONSHIP_2018
- HARTREE_KELVIN_RELATIONSHIP_2018
- HARTREE_KILOGRAM_RELATIONSHIP_2018
- HELION_ELECTRON_MASS_RATIO_2018
- HELION_G_FACTOR_2018
- HELION_MAG_MOM_2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- HELION_MASS_2018

- HELION_MASS_ENERGY_EQUIVALENT_2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- HELION_MASS_IN_U_2018
- HELION_MOLAR_MASS_2018
- HELION_PROTON_MASS_RATIO_2018
- HELION_RELATIVE_ATOMIC_MASS_2018
- HELION_SHIELDING_SHIFT_2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018
- HERTZ_HARTREE_RELATIONSHIP_2018
- HERTZ_INVERSE_METER_RELATIONSHIP_2018
- HERTZ_JOULE_RELATIONSHIP_2018
- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018
- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018
- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018
- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018

- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018
- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018
- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018
- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018
- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018
- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018

- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018
- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018
- NEUTRON_ELECTRON_MASS_RATIO_2018
- NEUTRON_G_FACTOR_2018
- NEUTRON_GYROMAG_RATIO_2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- NEUTRON_MAG_MOM_2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- NEUTRON_MASS_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_MASS_IN_U_2018
- NEUTRON_MOLAR_MASS_2018
- NEUTRON_MUON_MASS_RATIO_2018
- NEUTRON_PROTON_MAG_MOM_RATIO_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018
- NEUTRON_PROTON_MASS_RATIO_2018
- NEUTRON_RELATIVE_ATOMIC_MASS_2018
- NEUTRON_TAU_MASS_RATIO_2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018
- NUCLEAR_MAGNETON_2018
- NUCLEAR_MAGNETON_IN_EV_T_2018

- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- NUCLEAR_MAGNETON_IN_K_T_2018
- NUCLEAR_MAGNETON_IN_MHZ_T_2018
- PLANCK_CONSTANT_2018
- PLANCK_CONSTANT_IN_EV_HZ_2018
- PLANCK_LENGTH_2018
- PLANCK_MASS_2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018
- PLANCK_TEMPERATURE_2018
- PLANCK_TIME_2018
- PROTON_CHARGE_TO_MASS_QUOTIENT_2018
- PROTON_COMPTON_WAVELENGTH_2018
- PROTON_ELECTRON_MASS_RATIO_2018
- PROTON_G_FACTOR_2018
- PROTON_GYROMAG_RATIO_2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- PROTON_MAG_MOM_2018
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- PROTON_MAG_SHIELDING_CORRECTION_2018
- PROTON_MASS_2018
- PROTON_MASS_ENERGY_EQUIVALENT_2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- PROTON_MASS_IN_U_2018
- PROTON_MOLAR_MASS_2018
- PROTON_MUON_MASS_RATIO_2018
- PROTON_NEUTRON_MAG_MOM_RATIO_2018
- PROTON_NEUTRON_MASS_RATIO_2018
- PROTON_RELATIVE_ATOMIC_MASS_2018
- PROTON_RMS_CHARGE_RADIUS_2018
- PROTON_TAU_MASS_RATIO_2018
- QUANTUM_OF_CIRCULATION_2018
- QUANTUM_OF_CIRCULATION_TIMES_2_2018
- REDUCED_COMPTON_WAVELENGTH_2018
- REDUCED_MUON_COMPTON_WAVELENGTH_2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018
- REDUCED_PLANCK_CONSTANT_2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S_2018

- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018
- REDUCED_PROTON_COMPTON_WAVELENGTH_2018
- REDUCED_TAU_COMPTON_WAVELENGTH_2018
- RYDBERG_CONSTANT_2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2018
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018
- SECOND_RADIATION_CONSTANT_2018
- SHIELDED_HELION_GYROMAG_RATIO_2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018
- SPEED_OF_LIGHT_IN_VACUUM_2018
- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
- TAU_COMPTON_WAVELENGTH_2018
- TAU_ELECTRON_MASS_RATIO_2018
- TAU_ENERGY_EQUIVALENT_2018
- TAU_MASS_2018
- TAU_MASS_ENERGY_EQUIVALENT_2018
- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
- TAU_MUON_MASS_RATIO_2018
- TAU_NEUTRON_MASS_RATIO_2018

- TAU_PROTON_MASS_RATIO_2018
- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018
- TRITON_PROTON_MASS_RATIO_2018
- TRITON_RELATIVE_ATOMIC_MASS_2018
- TRITON_TO_PROTON_MAG_MOM_RATIO_2018
- UNIFIED_ATOMIC_MASS_UNIT_2018
- VACUUM_ELECTRIC_PERMITTIVITY_2018
- VACUUM_MAG_PERMEABILITY_2018
- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018
- W_TO_Z_MASS_RATIO_2018

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)